0.1 Topological Maps

Definition 0.1.1. Let X, Y be Topological Spaces. A map $f : X \to Y$ is said to be *continuous* at $x_0 \in X$ if:

For any open $V \in \mathcal{T}_Y$ with $f(x_0) \in V$, there is an open $U \in \mathcal{T}_X$ with $x_0 \in U$ such that $f(U) \subset V$

If f is continuous at every point in the domain, then f is called a *continuous map*.

Definition 0.1.2. Let X, Y be Topological Spaces.
A map $f : X \to Y$ is called an *open map* if: For any open $U \subseteq X$, $f[U] \subseteq Y$ is open in Y.
A map $f : X \to Y$ is called a *closed map* if: For any closed $C \subseteq X$, $f[C] \subseteq Y$ is closed in Y.

Theorem 0.1.3. Let $f : X \to Y$ be a map. Then, The Followings are Equivalent:

a) f is Continuous Map.

b) For any closed $C \subset Y$, $f^{-1}[C] \subset X$ be closed.

c) For any subset $A \subset X$, $f[\overline{A}] \subset \overline{f[A]}$.

d) For any subset $B \subset Y$, $f^{-1}[A^\circ] \subset (f^{-1}[B])^\circ$.

Proof.
a) $\implies$ **b)** Let $C \subset Y$ is closed. Then,

$$f^{-1}[Y \setminus C] = X \setminus f^{-1}[C]$$

is open in Y, thus $f^{-1}[C]$ is closed.
b) $\implies$ **c)** Let $A \subset X$ be given. Then,

$$A \subset f^{-1}[f[A]] \subset \underbrace{f^{-1}[\overline{f[A]}]}_{\text{closed by } b)} \implies \overline{A} \subset f^{-1}[\overline{f[A]}] \implies f[\overline{A}] \subset f[f^{-1}[\overline{f[A]}]] \subset \overline{f[A]}$$

c) $\implies$ **d)** Let $B \subset Y$, put $A = f^{-1}[Y \setminus B] = X \setminus f^{-1}[B]$. Then,

$$f[\overline{A}] = f[\overline{X \setminus f^{-1}[B]}] = f[X \setminus (f^{-1}[B])^\circ] \text{ and } \overline{f[A]} = \overline{f[f^{-1}[Y \setminus B]]} \subset \overline{Y \setminus B} = Y \setminus B^\circ$$

By c),

$$\begin{aligned} & f[\overline{A}] = f[X \setminus (f^{-1}[B])^\circ] \subset \overline{f[A]} \subset Y \setminus B^\circ \\ \implies & X \setminus (f^{-1}[B])^\circ \subset f^{-1}[f[X \setminus (f^{-1}[B])^\circ]] \subset f^{-1}[Y \setminus B^\circ] = X \setminus f^{-1}[B^\circ] \\ \implies & f^{-1}[B^\circ] \subset (f^{-1}[B])^\circ \end{aligned}$$

d) $\implies$ **a)** Let $U \subset Y$ be an open set. By d),

$$f^{-1}[U] \overset{U \text{ open}}{=} f^{-1}[U^\circ] \subset (f^{-1}[U])^\circ$$

Meanwhile, reverse inclusion is clear, thus $f^{-1}[U] = (f^{-1}[U])^\circ$, open. $\square$

Lemma 0.1.4. Let X, Y be Topological Space. Then,

1. $f : X \to Y$ is open map **if and only if** For any $A \subset X$, $f[A^\circ] \subset (f[A])^\circ$.
2. $f : X \to Y$ is closed map **if and only if** For any $A \subset X$, $\overline{f[A]} \subset f[\overline{A}]$.

Lemma 0.1.5. Let X, Y be Topological Space, and $f : X \to Y$ be a bijection. Then, TFAE:

1. f is open map.
2. f is closed map.
3. $f^{-1} : Y \to X$ be continuous map.

Clearly, Homeomorphism is open, closed, continuous map.

Lemma 0.1.6. Let $f : X \to Y$ be a Homeomorphism, $A \subset X$. Then, followings hold:

1. $f[\overline{A}] = \overline{f[A]}$.
2. $f[A^\circ] = (f[A])^\circ$.

Proof. 1. is clear by f is continuous, and closed map.
2. $\subset$) $A^\circ \subset A \implies f[A^\circ] \subset f[A] \implies f[A^\circ] \subset (f[A])^\circ \subset f[A]$ by f open map.
2. $\supset$) Let $x \in (f[A])^\circ$ be given. Then, there is an open $\mathcal{U} \in \mathcal{T}$ such that $x \in \mathcal{U} \subset f[A]$.
Now, $f^{-1}[x] \subset f^{-1}[\mathcal{U}] \subset f^{-1}[f[A]] = A$ by f is bijection, this implies
$(f^{-1}[\mathcal{U}])^\circ = f^{-1}[\mathcal{U}] \subset A^\circ \implies f^{-1}[x] \subset f^{-1}[\mathcal{U}] \subset A^\circ \implies f[f^{-1}[x]] = x \in f[A^\circ]$. ◻

Theorem 0.1.7. Let X, Y, Z be Topological Spaces, and $f : X \to Y$, $g : Y \to Z$ be functions.

1. If f, g are Continuous map, then $g \circ f$ is Continuous map.
2. If f, g are Open map, then $g \circ f$ is Open map.
3. If f, g are Closed map, then $g \circ f$ is Closed map.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & Y \arrow[r, "g"] & Z \end{tikzcd}$$

Above three statements are trivial.

1. If $g \circ f$ is Open map, f is Continuous onto map. Then, g is Open map.
2. If $g \circ f$ is Open map, g is Continuous one-to-one map. Then, f is Open map.

Proof. 1) Let $U \in \mathcal{T}_Y$ be an open set. Since f is Continuous map, $f^{-1}[U]$ is open of X.
Now,

$$\underbrace{(g \circ f)[\underbrace{f^{-1}[U]}_{\text{open}}]}_{\text{image of open map}} = g[f[f^{-1}[U]]] \overset{\text{onto}}{=} g[U] \in \mathcal{T}_Z$$

2) Let $U \in \mathcal{T}_X$ be an open set. Since $g \circ f$ is Open map, $(g \circ f)[U]$ is open of Z.
Now, by g is Continuous one-to-one,

$$g^{-1}[(g \circ f)[U]] = g^{-1}[g[f[U]]] \overset{1 \text{ to } 1}{=} f[U] \in \mathcal{T}_Y$$

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Lemma 0.1.8. *Pasting Lemma*
Suppose that X, Y are Topological Space, and $A, B \subset X$ such that $X = A \cup B$.
If both A, B are Open or Closed, $f : A \to Y$ and $g : B \to Y$ are Continuous map such that $f|_{A \cap B} = g|_{A \cap B}$, then

$$f \cup g : X \to Y : \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}$$

is Continuous map.